

# **CALARASI, Romania**

WMO: **154600**Lat: **44.2058N**Lon: **27.3383E**Elev: **65**StdP: **14.66**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999**

## **Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 9.9        | 15.0 | 5.1                           | 7.2 | 10.7 | 10.4 | 9.4 | 16.9 | 25.5                  | 26.1 | 21.7 | 26.0 | 5.5                   | 20   | 0.442 |

## **Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.5                   | 94.4            | 70.3 | 91.5 | 69.5 | 88.7 | 68.7 | 74.0                | 86.6 | 72.3 | 84.9 | 70.9 | 83.3 | 5.7                  | 90   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 70.3                            | 111.9 | 79.3 | 68.5 | 105.1 | 77.2 | 67.0 | 99.5 | 75.6 | 37.5          | 86.3 | 36.0 | 85.0 | 34.7 | 83.2 | 82.9              |

## **Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |  |  |
| Min               | Max  |      | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |  |  |
| 17.9              | 15.0 | 12.8 | 2.3                        | 99.0 | 5.8                | 3.9 | -1.9                                               | 101.8 | -5.3       | 104.1 | -8.5       | 106.3 | -12.7      | 109.2 |  |  |
|                   |      |      | 1.7                        | 76.4 | 5.5                | 2.8 | -2.2                                               | 78.4  | -5.5       | 80.0  | -8.6       | 81.6  | -12.6      | 83.6  |  |  |
|                   |      |      | DB                         |      |                    |     |                                                    |       |            |       |            |       |            |       |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |       |            |       |            |       |            |       |  |  |

## **Monthly Climatic Design Conditions**

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 54.6   | 31.5  | 36.3  | 44.9  | 54.5  | 64.0  | 71.6  | 75.3  | 74.7  | 66.0  | 54.8  | 45.1  | 34.7 |
|                                                                               | DBStd       | 16.94  | 9.42  | 9.99  | 8.34  | 6.80  | 6.10  | 5.44  | 4.69  | 5.00  | 6.29  | 7.62  | 8.32  | 8.58 |
|                                                                               | HDD50       | 1905   | 574   | 388   | 201   | 32    | 1     | 0     | 0     | 0     | 0     | 40    | 193   | 476  |
|                                                                               | HDD65       | 4816   | 1038  | 802   | 623   | 321   | 93    | 10    | 1     | 2     | 64    | 326   | 598   | 938  |
|                                                                               | CDD50       | 3567   | 1     | 6     | 43    | 166   | 434   | 647   | 786   | 767   | 481   | 188   | 45    | 3    |
|                                                                               | CDD65       | 1004   | 0     | 0     | 0     | 5     | 61    | 207   | 322   | 303   | 95    | 10    | 1     | 0    |
|                                                                               | CDH74       | 10620  | 0     | 0     | 6     | 100   | 699   | 2092  | 3394  | 3283  | 935   | 106   | 5     | 0    |
|                                                                               | CDH80       | 4362   | 0     | 0     | 0     | 18    | 163   | 816   | 1537  | 1516  | 295   | 17    | 0     | 0    |
| Wind                                                                          | WSAvg       | 5.3    | 6.0   | 6.3   | 6.6   | 6.0   | 5.2   | 4.7   | 4.5   | 4.4   | 4.9   | 4.7   | 5.1   | 5.7  |
| Precipitation                                                                 | PrecAvg     | 19.4   | 1.2   | 1.2   | 1.2   | 1.4   | 2.3   | 2.5   | 1.9   | 1.9   | 1.5   | 1.3   | 1.5   | 1.5  |
|                                                                               | PrecMax     | 31.9   | 5.8   | 3.4   | 3.8   | 2.5   | 6.8   | 5.2   | 4.5   | 5.7   | 4.7   | 5.5   | 4.6   | 5.9  |
|                                                                               | PrecMin     | 12.5   | 0.2   | 0.1   | 0.0   | 0.0   | 0.3   | 0.6   | 0.3   | 0.1   | 0.0   | 0.0   | 0.0   | 0.2  |
|                                                                               | PrecStd     | 4.4    | 1.1   | 0.7   | 0.9   | 0.6   | 1.5   | 1.3   | 1.1   | 1.4   | 1.3   | 1.1   | 1.1   | 1.2  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 57.3  | 65.8  | 73.3  | 82.5  | 88.2  | 95.1  | 99.5  | 98.7  | 91.1  | 81.9  | 73.3  | 61.5 |
|                                                                               |             | MCWB   | 49.9  | 53.4  | 54.8  | 62.4  | 66.5  | 71.5  | 70.0  | 70.3  | 66.8  | 64.7  | 60.1  | 53.4 |
|                                                                               | 2%          | DB     | 51.8  | 59.8  | 68.4  | 76.4  | 84.2  | 90.9  | 94.6  | 94.3  | 86.8  | 76.7  | 67.3  | 55.6 |
|                                                                               |             | MCWB   | 45.9  | 50.1  | 53.8  | 59.7  | 65.0  | 69.7  | 70.0  | 70.5  | 65.7  | 62.4  | 57.8  | 49.2 |
|                                                                               | 5%          | DB     | 47.5  | 54.5  | 64.1  | 71.8  | 81.1  | 87.9  | 91.4  | 91.4  | 83.3  | 72.1  | 62.0  | 50.4 |
|                                                                               |             | MCWB   | 43.6  | 47.3  | 51.8  | 57.6  | 64.2  | 69.1  | 69.8  | 69.6  | 64.9  | 60.3  | 55.3  | 46.4 |
|                                                                               | 10%         | DB     | 43.5  | 49.9  | 59.3  | 67.8  | 77.9  | 85.0  | 88.2  | 88.4  | 79.6  | 67.7  | 57.6  | 46.2 |
|                                                                               |             | MCWB   | 40.9  | 44.8  | 49.3  | 55.4  | 62.7  | 68.1  | 69.0  | 68.4  | 63.9  | 58.3  | 52.8  | 43.3 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 50.2  | 54.5  | 57.0  | 63.9  | 69.9  | 74.9  | 75.6  | 77.8  | 70.7  | 66.8  | 61.2  | 53.8 |
|                                                                               |             | MCDB   | 56.0  | 63.8  | 69.5  | 79.1  | 83.0  | 87.8  | 88.5  | 90.8  | 82.5  | 77.3  | 71.2  | 60.1 |
|                                                                               | 2%          | WB     | 46.7  | 50.8  | 54.5  | 61.1  | 67.6  | 72.8  | 73.7  | 74.3  | 68.4  | 64.1  | 58.5  | 50.0 |
|                                                                               |             | MCDB   | 50.9  | 58.3  | 66.2  | 74.6  | 80.0  | 85.9  | 86.7  | 86.0  | 79.4  | 73.4  | 65.7  | 54.7 |
|                                                                               | 5%          | WB     | 43.9  | 48.0  | 52.3  | 58.7  | 65.9  | 71.0  | 72.2  | 71.9  | 66.7  | 61.7  | 56.1  | 46.8 |
|                                                                               |             | MCDB   | 47.2  | 53.8  | 63.1  | 68.8  | 77.5  | 83.8  | 84.9  | 83.9  | 78.5  | 70.0  | 61.4  | 50.2 |
|                                                                               | 10%         | WB     | 41.1  | 44.9  | 49.9  | 56.7  | 64.0  | 69.5  | 70.9  | 70.2  | 65.2  | 59.3  | 53.3  | 43.5 |
|                                                                               |             | MCDB   | 43.9  | 49.6  | 58.1  | 65.5  | 74.7  | 81.5  | 83.2  | 83.0  | 76.7  | 66.4  | 57.4  | 46.3 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 11.6  | 14.7  | 17.9  | 20.9  | 22.2  | 21.9  | 22.5  | 23.8  | 21.9  | 18.9  | 14.6  | 12.0 |
|                                                                               |             | MCDBR  | 17.7  | 23.4  | 26.7  | 27.6  | 27.5  | 26.7  | 27.9  | 28.5  | 28.1  | 26.3  | 23.0  | 19.7 |
|                                                                               |             | MCWBR  | 12.6  | 14.8  | 14.1  | 13.6  | 12.1  | 9.7   | 9.0   | 9.4   | 10.6  | 13.1  | 13.8  | 14.1 |
|                                                                               | 5% WB       | MCDBR  | 17.1  | 22.6  | 25.1  | 24.1  | 24.4  | 23.9  | 23.4  | 23.9  | 24.6  | 23.8  | 21.3  | 18.7 |
|                                                                               |             | MCWBR  | 12.7  | 14.8  | 13.9  | 12.6  | 11.6  | 9.8   | 9.0   | 9.1   | 10.3  | 12.5  | 13.1  | 13.7 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.347  | 0.377 | 0.411 | 0.461 | 0.436 | 0.441 | 0.453 | 0.466 | 0.435 | 0.395 | 0.369 | 0.349 |      |
|                                                                               | taud        | 2.290  | 2.218 | 2.174 | 2.044 | 2.199 | 2.218 | 2.195 | 2.151 | 2.205 | 2.299 | 2.320 | 2.309 |      |
|                                                                               | Ebn at Noon | 238    | 251   | 259   | 254   | 266   | 265   | 259   | 251   | 248   | 242   | 228   | 224   |      |
|                                                                               | Edh at Noon | 27     | 35    | 41    | 51    | 45    | 44    | 45    | 45    | 40    | 32    | 26    | 24    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 438    | 690   | 1064  | 1499  | 1887  | 2048  | 2071  | 1853  | 1347  | 857   | 484   | 370   |      |
|                                                                               | RadStd      | 63     | 100   | 108   | 157   | 129   | 139   | 126   | 101   | 100   | 93    | 61    | 48    |      |

## **Historical Trends**

| Station Only            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Regional (48 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|                         | +0.85 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -214  | +157  | +99   |